

AI-Based Health Monitoring and Recommendation System

Abstract

The **AI-Based Health Monitoring and Recommendation System** is a rule-based artificial intelligence application developed using Python. The system collects user health parameters such as age, weight, height, gender, and activity level to analyze their health status. Based on this data, it calculates Body Mass Index (BMI), determines the user's health category, and provides personalized nutrition and lifestyle recommendations.

The system simulates intelligent decision-making by generating different outputs for underweight, normal, and overweight individuals. It also provides calorie intake or burn targets, health scores, and risk predictions, making it an effective AI-based assistant for basic health guidance.

Objectives

- To develop a system that calculates BMI and classifies health status
- To provide personalized diet and nutrition recommendations
- To simulate artificial intelligence using rule-based logic
- To generate calorie gain or loss plans based on user condition
- To provide health insights and risk analysis

Introduction to Artificial Intelligence in the Project

Artificial Intelligence (AI) refers to the ability of machines to simulate human intelligence and decision-making. In this project, AI is implemented using a **rule-based system**, where predefined conditions and logic are used to analyze input data and generate appropriate outputs.

The system behaves like an intelligent assistant by:

- Analyzing user health parameters
- Classifying users into health categories
- Providing personalized recommendations
- Generating health predictions

This approach is known as an **Expert System**, a branch of AI that uses logical rules instead of machine learning.

Methodology

The system follows these steps:

1. **Data Input**

User enters personal details such as age, weight, height, gender, and activity level.

2. **Data Processing**

The system calculates:

- BMI
- Ideal weight
- Macronutrient requirements
- Daily calorie needs

3. **Classification**

Based on BMI, the user is categorized into:

- Underweight
- Normal
- Overweight
- Obesity

4. **AI-Based Decision Making**

The system uses conditional logic to:

- Generate personalized diet or workout plans
- Provide calorie gain or burn targets
- Calculate health score
- Predict potential health risks

5. **Output Generation**

A complete health report along with AI recommendations is displayed.

Technologies Used

- Python Programming Language
- Conditional Statements (if-elif-else)
- Mathematical Calculations
- User Input/Output Handling

System Features

Basic Health Report

- BMI calculation and classification
- Ideal weight estimation
- Protein, carbohydrates, and fats requirement
- Daily calorie requirement
- Water intake recommendation
- Sleep duration suggestion
- Vitamins and minerals recommendation

AI-Based Features

- AI Health Score calculation
- Personalized weight gain plan for underweight users
- Personalized weight loss plan for overweight users
- Maintenance plan for normal users
- Daily calorie surplus/deficit targets
- Health risk prediction and insights

Input Parameters

The system requires the following inputs:

- Name
- Age
- Weight (in kilograms)
- Height (in meters)
- Gender (M/F/O)
- Activity level (sedentary/moderate/active/athlete)

Output

The system generates:

- Complete nutrition and health report
- BMI category and ideal weight
- Macronutrient requirements
- AI-based diet or workout plan
- Calorie gain or burn targets
- Health score

- Risk analysis

Working Principle

The system works on **rule-based AI logic**, where:

- If BMI < 18.5 → Underweight → Weight gain plan
- If BMI 18.5–25 → Normal → Maintenance plan
- If BMI > 25 → Overweight → Weight loss plan

This dynamic behavior makes the system adaptive and intelligent.

Advantages

- Easy to use and understand
- Provides instant health analysis
- Personalized recommendations
- No external data or internet required
- Demonstrates AI concepts clearly

Limitations

- Does not use real-time medical data
- Based on predefined rules only
- Not a substitute for professional medical advice
- No machine learning or advanced AI algorithms

Future Enhancements

- Integration with Machine Learning models
- Graphical User Interface (GUI)
- Mobile or web-based application
- Real-time health tracking
- Data storage and progress monitoring

Conclusion

The project successfully demonstrates how artificial intelligence concepts can be implemented using rule-based logic. It analyzes user data and provides intelligent recommendations, making it a functional AI-based health assistant.

Although it does not use advanced machine learning techniques, it effectively simulates AI behavior through decision-making and personalization. This makes it a valuable academic project for understanding the fundamentals of AI systems.

Developed By

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Project Type

AI-Based Academic Project (Rule-Based Expert System)